CFIFNet: A Road Extraction Network Through Cross-Modal Frequency-Domain Interaction and Fusion From High-Resolution Remote Sensing Image and GPS Trajectory/LiDAR

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***Abstract*—Recent advances in deep learning have significantly improved road extraction from high-resolution remote sensing (RS) images, yet challenges persist in scenarios with severe obstructions (e.g., forest canopies or dense fog) where semantic information is highly limited. To address these issues while overcoming the parameter inefficiency and suboptimal performance of existing multimodal approaches, we propose a lightweight cross-modal frequency-domain interaction and fusion network (CFIFNet) for effective interaction and fusion between high-resolution RS images and GPS trajectory/light detection and ranging (LiDAR) images based on the frequency domain. To facilitate accurate extraction of highly similar road-background edge regions, the model incorporates the wavelet interaction module (WIM), which uses the discrete wavelet transform (DWT) to decompose multimodal inputs into high-/low-frequency components—with high-frequency features undergoing global interaction via Transformer and low-frequency components undergoing local channel–spatial dual-branch interaction. In addition, the Fourier frequency-domain fusion (FFF) module enables efficient fusion through joint amplitude–phase manipulation, generating feature maps with enhanced semantic discrimination to improve detection of small roads and road-like objects. Moreover, the adaptive frequency-domain convolution attention (AFCA) is integrated to alleviate road edge occlusion by serially enhancing road features via dual-domain (channel and frequency) refinement. Evaluation metrics and visualization results on multiple multimodal road datasets validate CFIFNet's effectiveness, and a series of experiments demonstrate its efficiency in aligning cross-modal features, avoiding intermodal noise interference, and achieving complementary advantages in multisource road scenes.**

***Index Terms*—Adaptive frequency convolution attention-domain, Fourier frequency-domain fusion (FFF), wavelet interaction.**

I. Introduction

In RECENT years, the extraction of transportation roads has played a critical role in autonomous driving, automotive navigation systems, and smart cities. However, extracting roads from remote sensing (RS) images is more challenging compared with other RS targets because roads are typically thin and inconspicuous [1]. Influenced by advances in deep learning, road extraction from aerial RS images can be categorized into two main approaches. The first approach is based on vectorized road map structures, where the entire road topology network is extracted by iteratively predicting key points and the edges between them [2]. However, early stage keypoint prediction errors may propagate to subsequent connection steps, and the generalization capability for irregular roads (e.g., rural roads) remains insufficient. The second approach is based on pixel-level semantic segmentation to directly learn pixel-to-label mappings in an end-to-end manner, typically employing CNN-based methods including U-shaped encoder–decoder structures to predict road pixels [3], [4], while effectively capturing both global road features and road edge textures.

Zhou et al. [5] designed DLinkNet based on LinkNet, incorporating a multiscale dilated convolution module to expand the receptive field and learn multiscale road features, achieving good road extraction results on the DeepGlobe dataset. Furthermore, to enhance the identification of road features by learning long-range dependencies, Li et al. [6] developed spatial attention residual blocks to capture long-distance spatial dependencies and channel attention blocks to optimize multiscale feature fusion. Zhou et al. [7] employed graph convolution to enhance the expression of road features in both spatial and channel dimensions. With the rise of Transformers in the visual domain, studies [8], [9], [10], [11], [12], [13] have adopted Transformers and their variants as the backbone or skip connection branches to learn global features by capturing global dependencies. This approach addresses the issues of high road edge similarity and shadow occlusion

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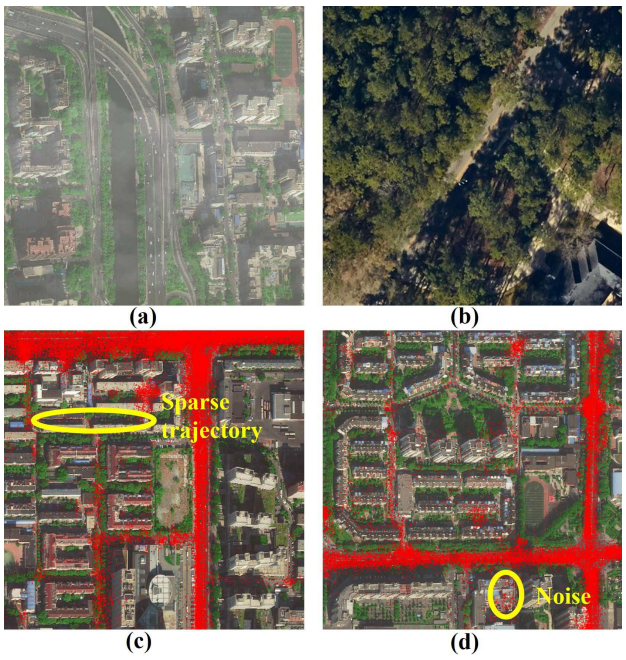


Fig. 1. Main limitations between aerial RS imagery and trajectories. (a) Dense fog scenario. (b) Forest canopies occlusion scenario. (c) Sparse GPS trajectory information in some primary roadway zones. (d) Trajectory points contain a significant amount of noise, including GPS track information in nonroad areas (e.g., parking lots).

by comparing the similarity of all pixels in the feature map. However, it also tends to learn redundant information unrelated to road features, significantly increasing computational burden.

Despite the high resolution as well as rich spatial and channel information provided by aerial RS images, they also have the limitation of complex backgrounds. For instance, rural roads may be covered by sand and shrubs, while urban road edges can be obscured by trees and buildings. Otherwise, in dense fog or forest canopies occlusion scenarios, high-resolution RS images may lack sufficient road information [shown in Fig. 1(a) and (b)], leading to a significant decline in model performance. Although numerous studies have attempted to address these issues through attention mechanisms and adaptive convolutions, their effectiveness in real-world applications remains limited. As a result, other modalities besides high-resolution RS images have been considered to overcome these challenges. Given the strong correlation between floating vehicle trajectory points and road distribution characteristics, many studies have adopted a multimodal fusion approach, combining RS images and floating vehicle trajectories for road extraction [14]. For example, some methods [15], [16] have fused the two data sources at the input stage along the channel dimension, using the same network structure as in single-modal approaches to extract road, but input-level fusion limits the integration and interaction of information between modalities, particularly when trajectory points are insufficient and sparse in major road regions as shown in Fig. 1(c). Liu et al. [1] proposed CMMPNet with a dual enhancement modules to refine the modal information representation of both aerial image and crowdsourced trajectories, thereby accurately extracting unconstrained roads, but road connectivity cannot be ensured due to discrete trajectory points collected at different time steps. Wang et al. [17] and Li et al. [18] have achieved road extraction in residential areas by fusing RS images and GPS trajectory features during the decoding phase of dual-modal branches. Wu et al. [19] proposed DeepDualMapper to implement coarse-to-fine road prediction through a novel deeply supervised decoder, with a gated fusion module (GFM) dynamically controlling information flow between modalities. However, trajectory information can also contain errors and deviations. For instance, trajectories in parking areas or some isolated points may interfere with accurate road identification, as illustrated in Fig. 1(d). Wang et al. [20] proposed a multistage feature fusion and regularization approach by integrating RS images, vehicle trajectories, and open-source maps to simultaneously extract road surfaces, centerlines, edges, and intersections. This method not only achieves accurate road segmentation but also effectively preserves road connectivity. While it can partially mitigate noise interference, the road vectorization process still heavily relies on the quality of the segmentation results.

Furthermore, since light detection and ranging (LiDAR) uses a highly penetrative laser to provide depth and distance information in top-down views, it is not affected by obstructions like buildings and trees. LiDAR can differentiate roads from other background features through laser reflections, making it widely used in multimodal RS fusion for object extraction. For instance, some studies [21], [22] have employed Transformer-based fusion of RS images and LiDAR images at the encoded feature layer to extract buildings. Method [1] has also used interleaved learning and information propagation approaches at the intermediate feature layer to extract roads by fusing RS images and LiDAR data. To minimize the impact of redundant features between modalities, Qiu et al. [14] designed a lightweight model CMIPNet with parameter-free and self-adaptive for road extraction. However, it faces challenges in complex urban street scenes due to an over-reliance on LiDAR information. Meanwhile, it is important to note that, similar to GPS trajectory maps, LiDAR feature maps also have limitations. For example, grayscale LiDAR maps may represent roads and other backgrounds, such as parking lots or parks, with similar grayscale values. This issue is even more pronounced in binary LiDAR maps, where large portions of the background may be confused with the foreground roads. Although LiDAR data can provide valuable supplementary information to high-resolution RS images, it can also introduce significant errors. Therefore, it is crucial to achieve effective complementarity and efficient fusion of high-resolution RS images and GPS trajectory/LiDAR images at the feature level, and to avoid learning redundant information.

To ensure effective road extraction while keeping the model parameters manageable, we designed a lightweight cross-modal frequency-domain interaction and fusion network (CFIFNet) composed of two independent encoders and a shared decoder. To facilitate efficient interaction between high-resolution RS images and GPS trajectory/LiDAR data during the encoding phase, we aim to enable mutual learning of beneficial information for enhancing road features while mitigating the impact and interference of redundant fea-

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tures. To achieve this, we introduced a wavelet interaction module (WIM) that interacts with low-frequency and high-frequency information of the two modalities, respectively, thereby obtaining consistent and distinctive features between them and promoting the refinement of road edge features. In addition, to provide the shared decoder with as much rich semantic information as possible and leverage the strengths of both modalities, we designed a Fourier frequency-domain fusion (FFF) module at the bottleneck position. It performs information fusion from the magnitude (containing global information) and phase (containing local information) to generate a semantically rich and high-contrast feature map. Finally, in the upsampling stage of the shared decoder, an adaptive frequency-domain convolution attention (AFCA) module is designed, which sequentially increase the discriminability between roads and backgrounds in occluded regions from both the channel dimension and frequency-domain perspective. In summary, our work can be summarized into four points.

1) We design a lightweight cross-modal fusion road extraction network, CFIFNet, using RS images and GPS trajectory/LiDAR images to complement the differences between modalities, with suppressing both intramodal and intermodal noise.
2) A WIM structure is designed within the model to facilitate the learning and complementarity of two modal information from the frequency domain, with aligned edge features of the two modalities.
3) A FFF module is developed to integrate the advantages of two modalities, fusing detailed semantic information, aiming to avoid the loss of small road and misclassification of objects with similar textures and colors to roads.
4) An AFCA module is designed to facilitate road features of two modalities to be efficiently provided to the feature recovery stage, with addressing the issue of shadow occlusion.

The following sections of this article are organized as follows. Section II provides an overview of related work on multimodal fusion for object extraction, frequency-domain filtering, and attention mechanism. Section III details our model's methodology. Section IV presents our experimental details and results, while Section V includes ablation experiments and analyses. Section VI is dedicated to the conclusion and outlook of this work.

II. Related Works

A. Multimodal Fusion for Object Extraction

Currently, there are three fusion strategies used for target extraction by combining multimodal data: input-level fusion, decision-level fusion, and feature-level fusion. Except in specific scenarios, feature-level fusion generally yields significantly better results than the other two methods in most tasks due to deep feature learning.

Liu et al. [23] employed a cross-modal adaptive fusion approach based on a generative adversarial network (GAN) for RGB-D salient object detection. This approach uses two encoder–decoder structures that separately process RGB and depth images as generators, while the fused result and ground-truth are fed into a discriminator. A cross-modal semantic segmentation model based on a Transformer was proposed by Ru et al. [24] for segmenting manufacturing workshops, enabling spatialwise and channelwise cross-modal information interaction between RGB and depth images. Similarly, Zhang et al. [25] proposed CMX to fuse RGB images with depth, LiDAR, thermal, and other features. However, in real-world scenarios, there may be some low-quality depth images due to issues such as camera shake, which can affect segmentation performance. Li et al. [26] proposed a multitask and multisource fusion network that enhances road detection performance by deeply integrating GPS trajectory information with RS images, with road intersections and road surface as the predicted outputs. Du et al. [27] combined hyperspectral images with high-resolution fused images to integrate deep semantic information, enhancing the perception of scene details in high-resolution images. An attention-based cross-fusion network was introduced by Chen et al. [28] for efficient building extraction by fusing multispectral information with normalized digital surface model (nDSM) data, but the extraction of some building boundaries is still not smooth enough. Liu et al. [29] designed a network SegFusion, which combined RS images with synthetic aperture radar (SAR) data for land cover classification. It employed a dual-hierarchical Transformer encoder to extract features from two modals at different stages, and used a Swin-Fusion module to integrate the above features at various levels. However, the multistage and multiscale Transformer operation significantly increases the computational cost. Duan et al. [30] designed a lightweight road extraction method based on GPS trajectory/LiDAR and RS images. Concurrently, a dynamic cross-modal information interaction mechanism was designed to make up for the redundant but useful information in the other modality, with a multimodal feature fusion module devised to tackle the interference of redundant noise.

Whilst the aforementioned methods have achieved promising results in the field of multisource fusion for RS feature extraction, they overly rely on spatial-domain feature interactions, which not only increases computational costs but may also lead to feature deviations during global feature interactions.

B. Frequency-Domain Filtering

Compared with the traditional Fourier transform, the wavelet transform can better process signals with instantaneous changes and automatically adjust the scale of analysis according to different characteristics of the signal.

Yang et al. [31] introduced a wavelet transform feature decomposer to decompose spatial features into low-frequency and high-frequency components. They also designed a dual-representation alignment filter to bridge the semantic gap between spatial and frequency-domain features, facilitating the integration of features across different representation domains. A Fourier convolutional neural network in the

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spatial–spectral domain was proposed by Liu et al. [32] for road segmentation. Its key components include a Fourier pooling layer in the encoding stage, asymmetric convolution, and squeeze–excitation inspired self-attention. Chen et al. [33] designed a frequency-domain adaptive dilation convolution to dynamically adjust the spatial dilation rate. This is accompanied by an adaptive convolutional kernel that decomposes convolution weights into low-frequency and high-frequency components, subsequently increasing the weight of the high-frequency part to capture more high-frequency information. Huang et al. [34] designed a frequency-domain decoupling bidirectional interactive network to address the issue of inaccurate target localization caused by the lack of internal texture and edge dispersion of the target. Influenced by the fact that high-frequency information contains detailed features within a local perspective and low-frequency information captures smooth variations within a global perspective, Qiao and Huang [35] proposed a frequency-domain feature extraction module to separately capture high-frequency and low-frequency features. An amplitude-guided phase-adaptive filter (AGPF) module was designed in the TSFNet [36] to explicitly disentangle and sequentially restore global common image degradation and specific structural information degradation in the frequency domain. Zhou et al. [37] designed a spatial–frequency information fusion network, which expands the receptive field of images through a modality-aware deep Fourier transform in the frequency-domain branch, with capturing global contextual information.

Even though the aforementioned methods learn contextual features and refine key information based on the frequency domain, efficient interaction of frequency-domain features and joint complementarity of frequency–spatial features remain an issue. Especially in the field of road extraction, how to fuse the high-frequency information containing road edges and the low-frequency information rich in global textures from RS images and GPS trajectories is an urgent problem that needs to be addressed.

C. Attention Mechanism

The attention mechanism is widely used in visual tasks, including RS object extraction and change detection, to enhance the distinctiveness of features and strengthen the connections between them by capturing long-range dependencies.

A dual attention dilated LinkNet was proposed by Gao et al. [38] to adaptively integrate local road features and global dependencies. A collaborative global–local network was proposed by Chen et al. [39] to efficiently preserve both global and local information, with the global branch processing the downsampled whole image while the local branch handles cropped local patches. Li et al. [40] introduced a dense skip connection module and a hybrid attention module to fully utilize contextual information and extract discriminative features. Cao et al. [41] proposed a cross-scale axial attention mechanism, which efficiently integrates high-resolution fine-grained features and low-resolution coarse-grained features by sequentially applying axial attention in the horizontal and vertical directions. A position attention mechanism was designed in TransRoadNet [42] to enhance the expressiveness of road features, accompanied by a contextual information extraction module which was meticulously designed to capture comprehensive road feature context information. In SAGNet [43], a global semantic aggregation module was proposed to extract semantic change information, and an attention fusion module was employed to refine features for change map reconstruction. Hou et al. [44] proposed a position attention that captures long-range dependencies along the spatial direction while embedding the position information into other spatial directions. The resulting direction-aware attention map and position-sensitive attention map can complementarily enhance the target representation. Wang et al. [45] employed an efficient global–local attention mechanism combined with a Transformer and CNN to model both global and local information.

Although extensive studies have demonstrated attention's effectiveness in long-range modeling, applying attention mechanisms in the interaction and fusion of multisource data remains a challenge.

III. Methods

A. Main Framework

In this article, our CFIFNet consists of two encoder branches and one common decoder. The encoders are used for mapping and downsampling inputs of the two modalities. The encoder branch that takes high-resolution RS images as input uses ResNet-34 as the backbone, while the branch that processes GPS trajectory maps/LiDAR feature maps uses a simpler architecture consisting of 3×3 convolution + pooling + batch normalization + ReLU activation. This design is due to the fact that high-resolution RS images contain richer semantic information, and the features in the channel dimension are more discriminative, thus requiring deeper feature extraction and learning. In contrast, the GPS trajectory/LiDAR maps, once processed into image format, are single-channel grayscale images, and using overly complex CNN structures may lead to overfitting. Meanwhile, each encoder incorporates a dBlock structure at its end—responsible for expanding the receptive field and capturing multiscale contextual information—while the shared decoder handles upsampling following the fusion of the two modalities, and these components collectively constitute the baseline model. Moreover, to ensure sufficient complementarity and interaction between the two modalities during the encoding and dBlock phases, we designed the WIM that enables cross learning of the two modalities and the FFF that promotes effective fusion in the frequency domain. In addition, the AFCA is designed to enhance road features based on channel weighting and frequency-domain attention, with the WIM and AFCA, respectively, incorporated into every downsampling and upsampling stage. The overall CFIFNet is shown in Fig. 2.

B. Wavelet Interaction Module

In this study, a WIM is designed to facilitate the interactive learning of multisource data. First, the Haar discrete wavelet transform (DWT) is applied separately to high-resolution RS feature maps and GPS trajectory feature maps to extract their

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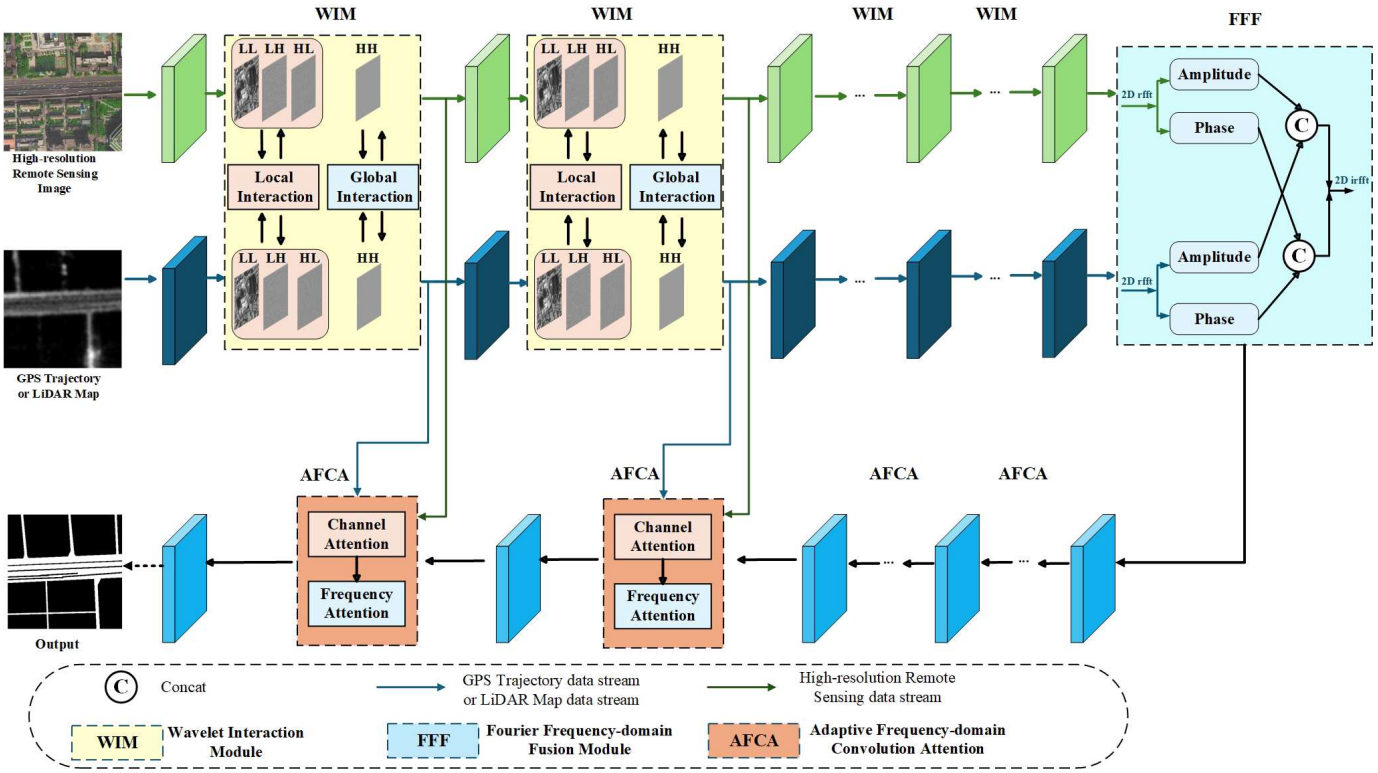


Fig. 2. Structural overview diagram of the CFIFNet.

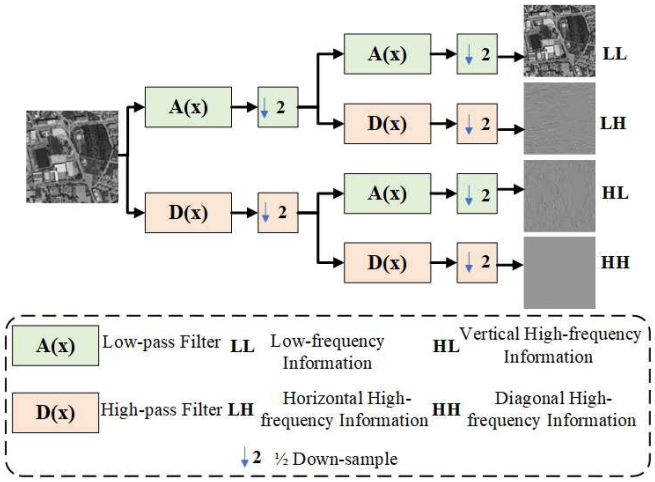


Fig. 3. Process of Haar wavelet transform.

frequency-domain features. The process of the Haar wavelet transform is shown in Fig. 3.

Here, $A(x)$ and $D(x)$ represent the low-pass and high-pass filters, respectively, while $\downarrow 2$ indicates the downsampling to half the original size. Continuous application of low-pass or high-pass filters is required to obtain the final output.

Specifically, high-resolution RS features are first decomposed along the row direction of F_I using low-pass and high-pass filters, respectively, yielding the wavelet subbands I_L and I_H. Subsequently, the same low-pass and high-pass filters are applied along the column direction to further decompose the results, eventually obtaining the subbands I_{LL}, I_{LH}, I_{HL}, and I_{HH}. Among these, I_{LL}, I_{LH}, I_{HL}, and I_{HH} represent low-frequency information, horizontal high-frequency information, vertical high-frequency information, and diagonal high-frequency information, respectively. Similarly, GPS trajectory features are also processed using the DWT, resulting in the wavelet subbands T_{LL}, T_{LH}, T_{HL}, and T_{HH}. The concatenated subbands are denoted as $I = C[I_{LL}, I_{LH}, I_{HL}, I_{HH}]$ and $T = C[T_{LL}, T_{LH}, T_{HL}, T_{HH}]$, where C denotes the concatenation along the channel dimension. These subbands are then processed for interaction in high and low frequencies, respectively. The Haar wavelet transform process is mathematically described in the following equations:

$$A_c(i,j) = \frac{X_c(i,2j-1) + X_c(i,2j)}{2} \tag{1}$$

$$D_c(i,j) = X_c(2i-1,j) - X_c(2i,j) \tag{2}$$

$$LL_c(i,j) = AA_c(i,j) = \frac{A_c(i,2j-1) + A_c(i,2j)}{2} \tag{3}$$

$$LH_c(i,j) = AD_c(i,j) = \frac{D_c(i,2j-1) + D_c(i,2j)}{2} \tag{4}$$

$$HL_c(i,j) = DA_c(i,j) = A_c(i,2j-1) - A_c(i,2j) \tag{5}$$

$$HH_c(i,j) = DD_c(i,j) = D_c(i,2j-1) - D_c(i,2j). \tag{6}$$

Here, A_c and D_c represent the low-pass and high-pass filters for each channel, i is the row index ranging from 1 to $H/2$, and j is the column index ranging from 1 to $W/2$.

1) Local Interaction: For the low-frequency components of the two modalities, i.e., I_{LL} and T_{LL}, local interactive learning and complementarity are performed in both channel and spatial dimensions. First, in channel dimension, I_{LL} and T_{LL} are added to obtain the initial fused feature map F. The F is subjected to global average pooling (GAP) in the spatial

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dimensions (height and width) to generate channel weighting coefficients C_w. Then, C_w is multiplied by the original I_{LL} and T_{LL} to obtain I_C and T_C, enabling features from one modality to learn from the other modality, thereby enhancing their channel-level differences. Next, I_C and T_C are added to obtain consistent fused features C_{fuse}, while their subtraction yields modality-specific difference features C_{diff}. Afterward, in spatial dimension, the F undergoes GAP and maximum pooling in the channel dimension to obtain S_{avg} and S_{max}. These are concatenated along the channel dimension and passed through a 3×3 convolution to generate spatial attention weighting coefficients S_w. Multiplying S_w with I and T yields the weighted feature maps I_S and T_S, respectively, which are then added to produce consistent fused features S_{fuse} and subtracted to generate difference features S_{diff}. The consistent fused features $[C_{\text{fuse}}, S_{\text{fuse}}]$ and difference features $[C_{\text{diff}}, S_{\text{diff}}]$ from the channel and spatial dimensions are then combined to produce F_{fd}, where $F_{\text{fd}} = C_{\text{fuse}}+C_{\text{diff}}+S_{\text{fuse}}+S_{\text{diff}}$. This F_{fd} is added elementwise to the original I_{LL} and T_{LL}, resulting in I_{low} and T_{low}, which integrate complementary information from the other modality, enriching the intramodality information.

2) Global Interaction: For the high-frequency components, i.e., I_{LH}, I_{HL}, and I_{HH} and T_{LH}, T_{HL}, and T_{HH}, these are, respectively, concatenated along the channel dimension to obtain I_{high} and T_{high}. Following this, we perform linear mappings on I_{high} and T_{high} to obtain Q_I, K_I, and V_I and Q_T, K_T, and V_T, respectively. Then, we conduct multihead Transformer computations on Q_I, K_T, and V_T, and also on Q_T, K_I, and V_I, ensuring that high-resolution RS high-frequency features learn global information from GPS trajectory high-frequency features, while GPS trajectory high-frequency features learn from high-resolution RS high-frequency features. Refer to (7)–(10), where d and D denote the scaling factor and bias matrix, respectively. In addition, W_{qI}, W_{kI}, and W_{vI} represent the trainable weight matrices of Q_I, K_I, and V_I, while W_{qT}, W_{kT}, and W_{vT} represent that of Q_T, K_T, and V_T

$$Q_I = W_{qI}I_{\text{high}}, K_I = W_{kI}I_{\text{high}}, V_I = W_{vI}I_{\text{high}} \tag{7}$$

$$Q_T = W_{qT}T_{\text{high}}, K_T = W_{kT}T_{\text{high}}, V_T = W_{vT}T_{\text{high}} \tag{8}$$

$$I'_{\text{high}} = \text{Softmax}\left(\frac{Q_I K_T}{\sqrt{d}} + D\right)V_T \tag{9}$$

$$T'_{\text{high}} = \text{Softmax}\left(\frac{Q_T K_I}{\sqrt{d}} + D\right)V_I. \tag{10}$$

This promotes global complementarity in high-frequency information across the modalities. Then, I'_{high} is concatenated with I_{low} along the channel dimension to form I', and T'_{high} is concatenated with T_{low} to form T'.

Finally, the updated frequency-domain features I' and T' are subjected to the inverse DWT (IDWT) to obtain their corresponding updated spatial-domain feature maps F_I and F_T. Through effective interaction of low-frequency and multidirectional high-frequency information in the frequency domain, the high-frequency features of both modalities (including road edge features) are further enriched and refined. The structure of the WIM module is shown in Fig. 4.

C. FFF Module

The 2-D fast Fourier transform (2D-FFT) is a method for converting a 2-D spatial signal (e.g., an image) from the spatial domain to the frequency domain [46]. Through this transformation, this method can represent the pixel values of an image as a combination of different frequency components, enabling a clearer representation of the structural information of the image in the frequency domain. Specifically, low-frequency information is concentrated at the center of the spectrum, representing the global structure and contour information of the image, while high-frequency information is located at the edges of the spectrum, containing details, edges, and noise.

The FFF adopts the 2-D real fast Fourier transform (rfft2) function to perform the Fourier transform, which leverages the symmetry of the Fourier Transform to reduce computational complexity. Its output is a complex matrix but contains only half of the spectral information because, for real-valued input signals, the Fourier Transform exhibits conjugate symmetry. Compared with the conventional 2D-FFT, rfft2 is computationally more efficient, especially for large-scale data, as it only calculates and stores half of the frequency components.

After the two encoding branches perform interaction on the data from two modalities, this method applies the rfft2 operation to the high-resolution RS features F_I and GPS trajectory features F_T in the bottleneck layer of the encoder–decoder structure to obtain the frequency-domain features I_{freq} and T_{freq} of the two modalities. Then, the torch.abs function and torch.angle function are used to extract the amplitude I_{amp} and phase I_{pha} corresponding to the frequency-domain feature I_{freq}, as well as the amplitude T_{amp} and phase T_{pha} corresponding to T_{freq}.

Next, the amplitudes I_{amp} and T_{amp} of the two modalities are concatenated along the channel dimension and fused via a 1×1 convolution operation to produce the fused amplitude F_{amp}. Similarly, the phases I_{pha} and T_{pha} are concatenated along the channel dimension and fused via another 1×1 convolution to generate the fused phase F_{pha}. Subsequently, this study constructs the real part (real) and imaginary part (imag) of the fused frequency-domain feature map, as shown in the following equation:

$$\text{real} = F_{\text{amp}} * \cos\left(F_{\text{pha}}\right), \quad \text{imag} = F_{\text{amp}} * \sin\left(F_{\text{pha}}\right). \tag{11}$$

Finally, the frequency-domain features [real, imag], represented as a complex matrix, are transformed back to the spatial domain using the 2-D inverse real fast Fourier transform (irfft2) to obtain the fused spatial-domain feature maps. The structure of the FFF module is shown in Fig. 5.

D. Adaptive Frequency-Domain Convolution Attention

Finally, to address potential information loss during the upsampling process, this study designs the AFCA as a skip connection to enhance road features during the decoding stage. Due to significant differences in channel features between the encoding and decoding stages, it is crucial to align the feature information provided by the encoding stage with the channel characteristics of the feature maps in the decoding stage.

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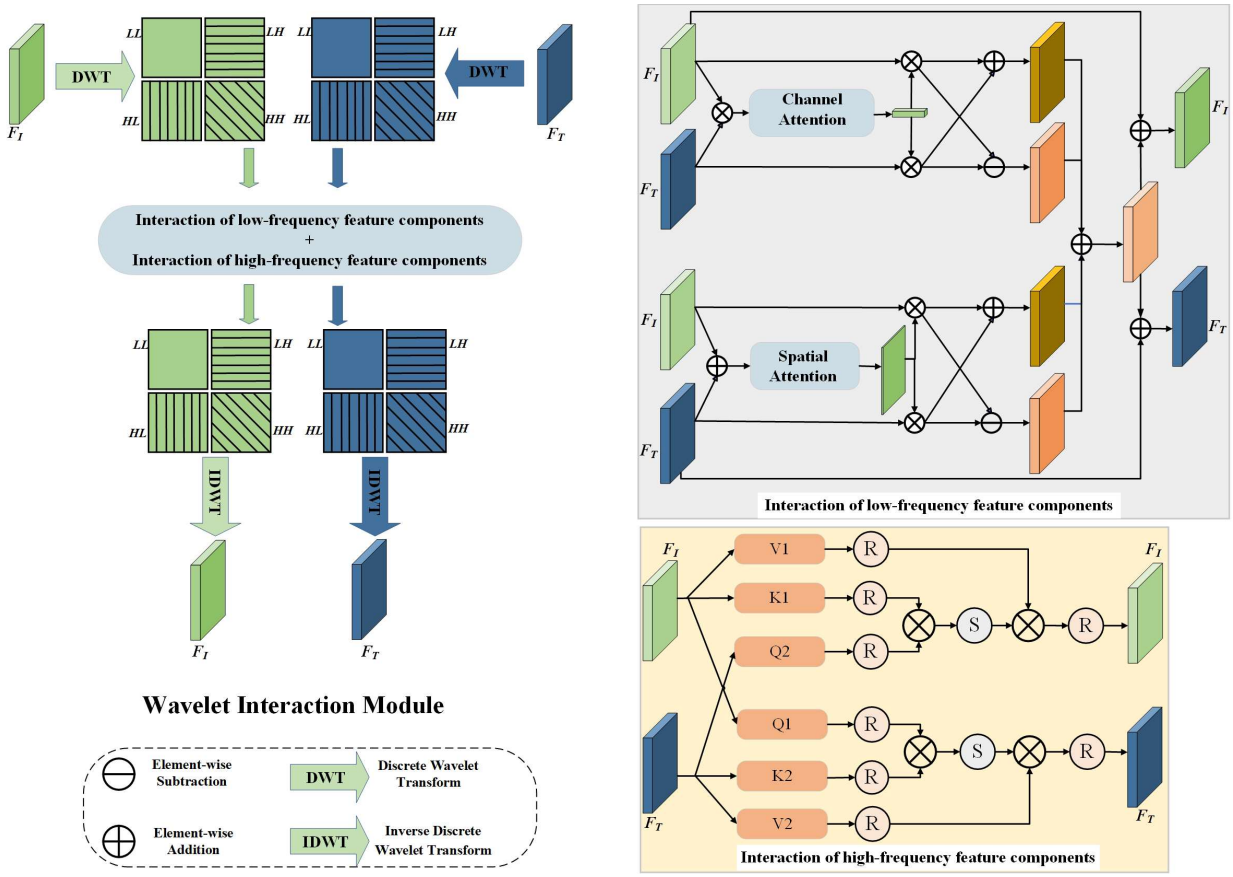


Fig. 4. Detailed structure of the WIM.

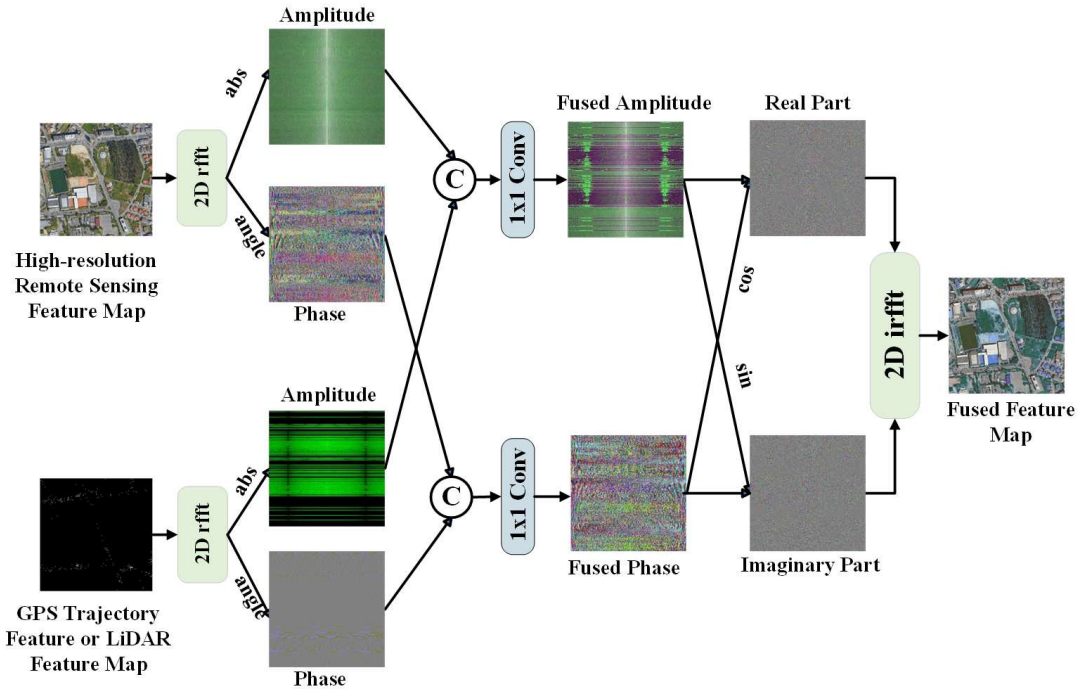


Fig. 5. Detailed structure of FFF module.

Specifically, this module first concatenates high-resolution RS features with GPS trajectory features along the channel dimension. Then, a channel attention layer is applied to the channel features of the two modalities to assign different

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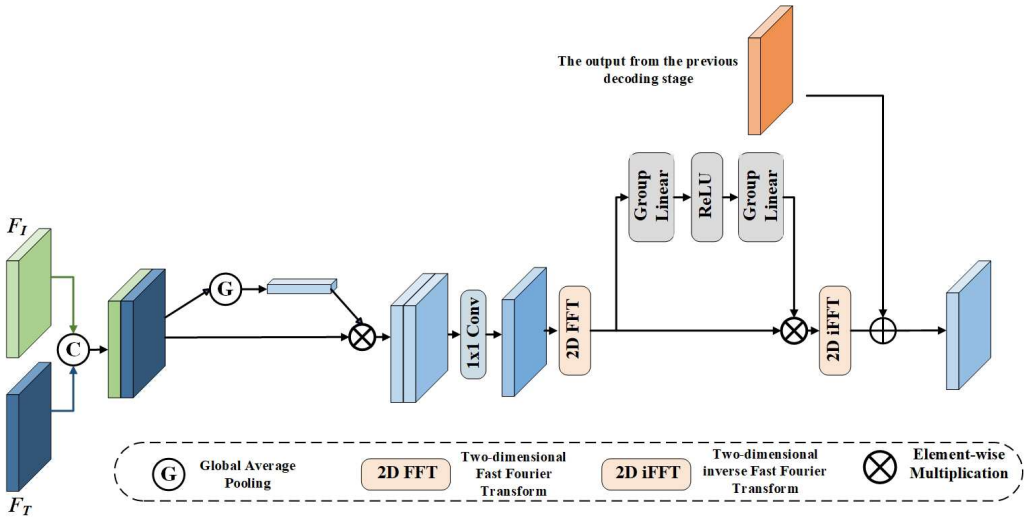


Fig. 6. Detailed structure of AFCA module.

weights, enhancing the distinction between different feature categories and highlighting road features.

After the fusion of modal data via a 1×1 convolution along the channel dimension, adaptive frequency-domain filtering is performed, with the filter operator generated from the Fourier-transformed features. Moreover, the matrix multiplication between the filter operator and the frequency-domain feature map is equivalent to a large-size kernel convolution in the spatial domain, which further expands the receptive field.

Finally, the resulting feature map is added pixelwise to the output from the previous decoding stage to generate the final output of this module. In addition, the method does not employ overly complex techniques, as such methods can lead to overfitting and slow down the training process. The adaptive frequency-domain filtering further reduces the number of parameters when learning neighboring and remote features. The structure of the AFCA module is shown in Fig. 6.

E. Loss Function

Since our road extraction task is based on semantic segmentation, and considering the extreme foreground–background imbalance in road scenes, we utilize the FocalLoss function to address the pixel-level classification challenge. In addition, we employ DiceLoss to focus on the minimal foreground information from a more global perspective [13], as shown in (12) and (13). Here, t and p represent the ground-truth pixel and the predicted pixel, respectively, while s denotes the smoothing coefficient set to $1e^{-8}$

$$L_f = -\alpha (1 - p_i)^{\gamma} \log (p_i) \tag{12}$$

$$L_d = 1 - \frac{\sum_{i=1}^{n} p_i t_i + s}{\sum_{i=1}^{n} p_i + \sum_{i=1}^{n} t_i + s}. \tag{13}$$

IV. Experiments

A. Road Dataset

1) BJRoad Dataset: The BJRoad dataset [15] consists of two modalities of data sampled in the Beijing area of China: high-resolution RS images and GPS trajectory maps. The dataset contains 1350 high-resolution RS images and 50 million trajectory points collected from floating vehicles. These trajectory points are processed through counting and filtering to generate grayscale trajectory maps corresponding to the 1350 high-resolution RS images. Each image has a size of 1024×1024 pixels and a resolution of 0.5 m/pixel. For training and testing our model, we divided the entire dataset into 70% for the training set, 20% for the testing set, and 10% for the validation set.

2) TLCGIS Dataset: The TLCGIS dataset [47] consists of two modalities of data: high-resolution aerial RS images and LiDAR images, with 5860 images in each modality. Each image has a size of 500×500 pixels with a resolution of 0.5 m/pixel. Notably, in the official ground-truth labels of this dataset, road pixels are labeled as 0 and background pixels as 1. However, in our experiments, we converted the road pixels to 1 and the background pixels to 0. Finally, we divided the entire dataset into 2640 images for the training set, 2400 images for the testing set, and 240 images for the validation set.

3) Porto Dataset: The Porto dataset [19] covers an area of approximately 209 km^2 in the Porto area of Portugal. It consists of 6048 pairs of high-resolution RS images and GPS trajectory maps, each with a size of 512×512 pixels. The trajectory maps are derived from the trajectory coordinate data collected from 442 taxis in these regions during 2013 and 2014. Due to the lack of a designated test set in the original dataset, we divided the entire dataset into five parts, using four-fifths for the training set and the remaining one-fifth for the testing set.

B. Experimental Details

All experiments were conducted on two NVIDIA GeForce RTX 3050 with 12-GB memory each, using the PyTorch framework and Python 3.10 in a Linux environment. The Adam optimizer is used to update the model parameters. The batch size is set to 4, the initial learning rate is set to $2e^{-4}$,

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and a poly learning rate adjustment strategy is employed to dynamically adjust the learning rate as the number of epochs increases. Moreover, the total number of epochs is set to 30 for the BJRoad dataset and 50 for the TLCGIS and Porto datasets, due to the larger sample size of the TLCGIS and Porto datasets. Finally, an early stopping strategy is employed during training to prevent overfitting. Specifically, if the performance on the validation set falls below the previous best value for five consecutive epochs, the training process is terminated early.

C. Evaluation Indicators

Since our task involves road extraction based on pixel-level classification, we use the $F1$-score ($F1$) and intersection over union (IoU) as evaluation metrics to assess the effectiveness of the CFIFNet, as shown in the following equations:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$F1 = \frac{2P \cdot R}{P + R} \tag{16}$$

$$\mathrm{IoU} = \frac{TP}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}}. \tag{17}$$

Here, TP represents the number of road pixels correctly identified as road, and FP represents the number of background pixels incorrectly identified as road. TN represents the number of background pixels correctly identified as background, and FN represents the number of road pixels incorrectly identified as background [7].

Moreover, we also introduced the metric, average path length similarity (APLS) [48], to assess the connectivity of road extraction. Specifically, the APLS calculates the sum of the optimal path length differences between nodes in the ground-truth graph G and the predicted graph G'. The APLS ranges from $[0, 1]$, with higher values indicating better performance. The APLS can be computed using the following equations:

$$S_{P->T}\left(G, G'\right) = 1 - \frac{1}{M} \sum \min\left(1, \frac{|L(a,b) - L(a',b')|}{L(a,b)}\right) \tag{18}$$

$$S_{T->P}\left(G', G\right) = 1 - \frac{1}{M} \sum \min\left(1, \frac{|L(a,b) - L(a',b')|}{L(a',b')}\right) \tag{19}$$

$$\mathrm{APLS} = \frac{1}{N} \sum \frac{1}{\frac{1}{S_{P->T}} + \frac{1}{S_{T->P}}} \tag{20}$$

where M and N denote the number of unique paths and the number of images, respectively. Nodes a' and b' represent the nodes in G' that are closest to the ground-truth nodes a and b in location. $L(a, b)$ and $L(a', b')$ denote the shortest path lengths between $a \to b$ in G and $a' \to b'$ in G', respectively. $S_{P \to T}$ measures the total shortest path differences between each pair of nodes in G and G'. The definition of $S_{T \to P}$ is similar. For missing nodes with high centrality, the APLS imposes a heavier penalty compared to missing nodes with low centrality. Therefore, a high score can only be achieved when the roads are correctly connected.

TABLE I

COMPARISON OF PERFORMANCE AMONG DIFFERENT MODELS IN BJROAD DATASET, WHERE BOLD NUMBERS INDICATE THE OPTIMAL VALUE FOR THIS METRIC

	IoU	F1	APLS
Ours	**0.650**	**0.787**	**0.592**
DEFNet (TGRS'2025)	0.606	0.748	0.532
CMMPNet (TNNLS'2022)	0.634	0.773	0.543
CMIPNet (TGRS'2024)	0.640	-	0.587
SGACNet (Sensors'2023)	0.627	0.769	0.571
CMFormer (PR'2023)	0.629	0.771	0.533
CMX (TITS'2023)	0.623	0.766	0.583
DLinkNet (CVPR'2018)	0.580	0.729	0.499
Sun et al. (CVPR'2019)	0.592	0.743	0.558
SA-Gate (ECCV'2020)	0.626	0.769	0.578

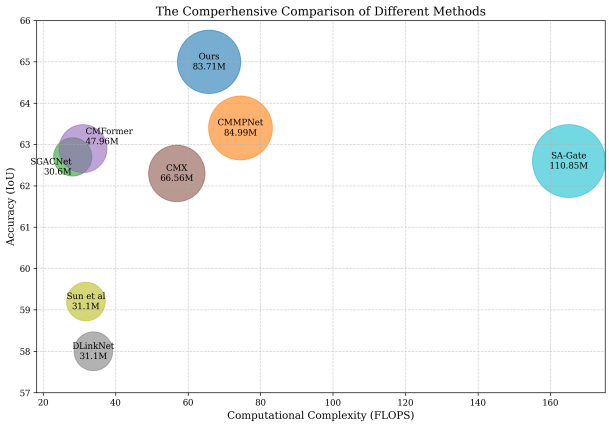


Fig. 7. Comprehensive comparison of accuracy (IoU), computational complexity (FLOPs), and parameter count (Params), where larger areas of each bubble indicate more complex models.

D. Results

In the comparative experiments, we compare and analyze CFIFNet against models specifically designed for multimodal road extraction, including CMMPNet [1], CMIPNet [14], DEFNet [49], and the method proposed by Sun et al. [15], as well as widely used multimodal semantic segmentation models such as CMFormer [24], CMX [25], SGACNet [50], and SA-Gate [51].

1) Experimental Results on the BJRoad Dataset: In Table I, the model of this study achieved the best performance in all metrics, demonstrating its ability to leverage the advantages between modalities while effectively suppressing noise between them. Specifically, CFIFNet surpasses the feature-level fusion multimodal models CMMPNet and CMIPNet in the IoU metric by 1.0% and 1.6%, respectively. Furthermore, it outperforms the input-level fusion model proposed by Sun et al. [15] by 5.8% in IoU. This excellent performance can be attributed to the limited ability of these models to capture and interact with multisource information, as they only perform simple channel connections at the input stage. It can also be observed that CFIFNet has a higher APLS value than all other comparative models, outperforming the parameter-free and adaptive road extraction model CMIPNet by 0.5%, which indicates that CFIFNet can extract more connected roads and more complete topological structures.

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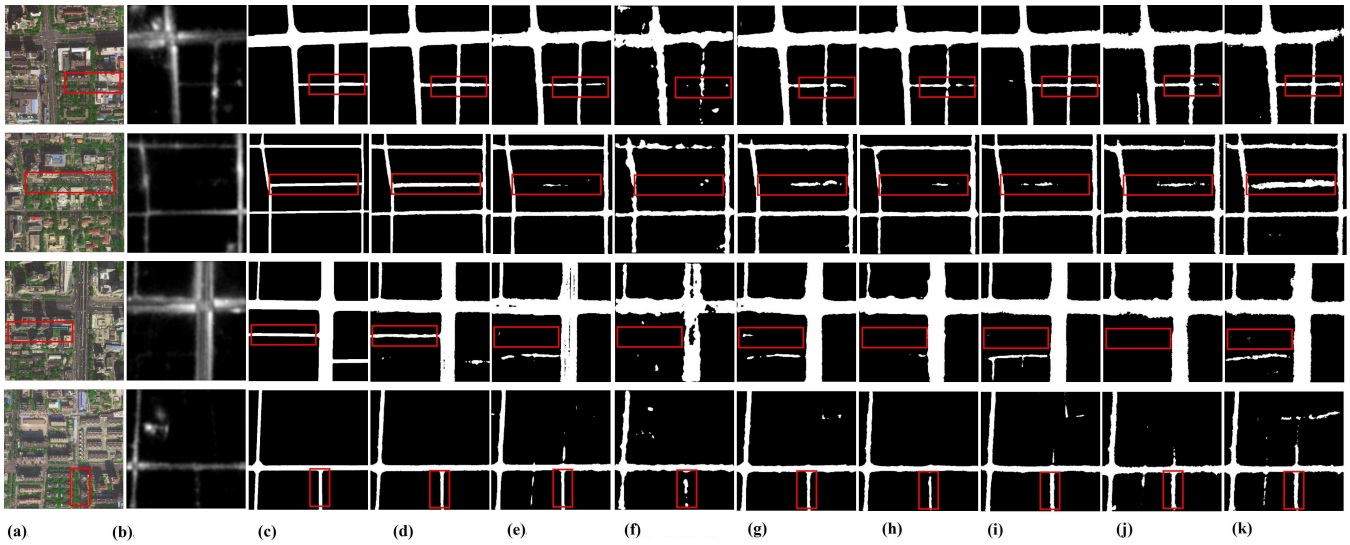


Fig. 8. Comparison of visual results among models on the BJRoad dataset. (a) High-resolution RS. (b) GPS trajectory. (c) Ground truth. (d) Ours. (e) CMXNet. (f) SGACNet. (g) CMFormer. (h) SA-Gate. (i) CMMPNet. (j) Sun et al. (k) DLinkNet.

TABLE II
COMPARISON OF PARAMS AND FLOPS AMONG DIFFERENT MODELS IN BJROAD DATASET, WHERE BOLD NUMBERS INDICATE THE OPTIMAL VALUE FOR THIS METRIC

	Params(M)	Flops(G)
Ours	83.71	65.81
CMMPNet	84.99	74.48
CMIPNet	-	-
SGACNet	30.6	28.17
CMFormer	47.96	31.01
CMX	66.56	56.90
DLinkNet	31.1	33.88
Sun et al.	**31.1**	**31.81**
SA-Gate	110.85	165.08

In addition, from Table II and Fig. 7, it can be seen that our CFIFNet achieves an optimal performance balance between accuracy and computational complexity. Specifically, the number of parameters and floating-point operations or model inference time of the model are lower than those of CMMPNet and SA-Gate. This is due to the CFIFNet enhancing features and suppressing noise based on the frequency domain when interacting and fusing information from the two modalities, represented by WIM, FFF, and AFCA. In contrast with it, CMMPNet and SA-Gate, which are based on spatial convolution and attention-based information interaction and fusion, inevitably increase the number of parameters and model inference time. Moreover, although the number of parameters in CFIFNet is slightly higher than that of cross-modal fusion semantic segmentation models such as SGACNet and CMFormer, the study still achieves superior performance in road extraction.

In Fig. 8, when urban street roads are obscured by the shadows of trees and buildings, CFIFNet is capable of accurately identifying road edge areas. This effectiveness is attributed to the efficient integration and effective supplementation of high-resolution RS features and GPS trajectory information, primarily due to the separate interaction of high-frequency and low-frequency information in the WIM module, which, to some extent, promotes attention to edge areas. Furthermore, it can be observed that other models tend to predict certain street areas that resemble roads in shape, despite these streets being unsuitable for vehicle traffic and lacking GPS trajectory information. Due to insufficient utilization of trajectory information, these models predict false positive samples. In addition, as seen in the fourth row, the inclusion of GPS trajectories leads other models to easily predict areas rich in GPS trajectory information, but some of these areas may merely be parking lots, which, despite having a large number of trajectories, ultimately result in false negative predictions. In contrast, our method utilizes high-resolution RS data and GPS trajectory information, with comprehensive integration from the perspective of amplitude and phase values in the frequency domain in the FFF module, further suppressing noise interference and enhancing the difference between roads and background. Meanwhile, the channel weighting and frequency adaptive filtering in the AFCA module also provide rich road information during the data reconstruction stage, thus ultimately achieving accurate road recognition. In the third row, other models overly rely on high-resolution RS information or GPS trajectory data and fail to leverage the complementary advantages of both modalities, leading to incorrect predictions of road locations.

2) Experimental Results on the Porto Dataset: In Table III, our CFIFNet once again achieves state-of-the-art (SOTA) performance, surpassing the second-place model CMMPNet by 2.2% in IoU. In addition, it outperforms early fusion models DLinkNet and Sun et al. by 5.3% and 6.7% in IoU, respectively. This demonstrates the effectiveness of the CFIFNet in utilizing multisource data for road extraction. It also indicates that our model can fully leverage the characteristics of GPS trajectory maps, which are unaffected by weather conditions and complex urban scenes, while effectively aligning the road

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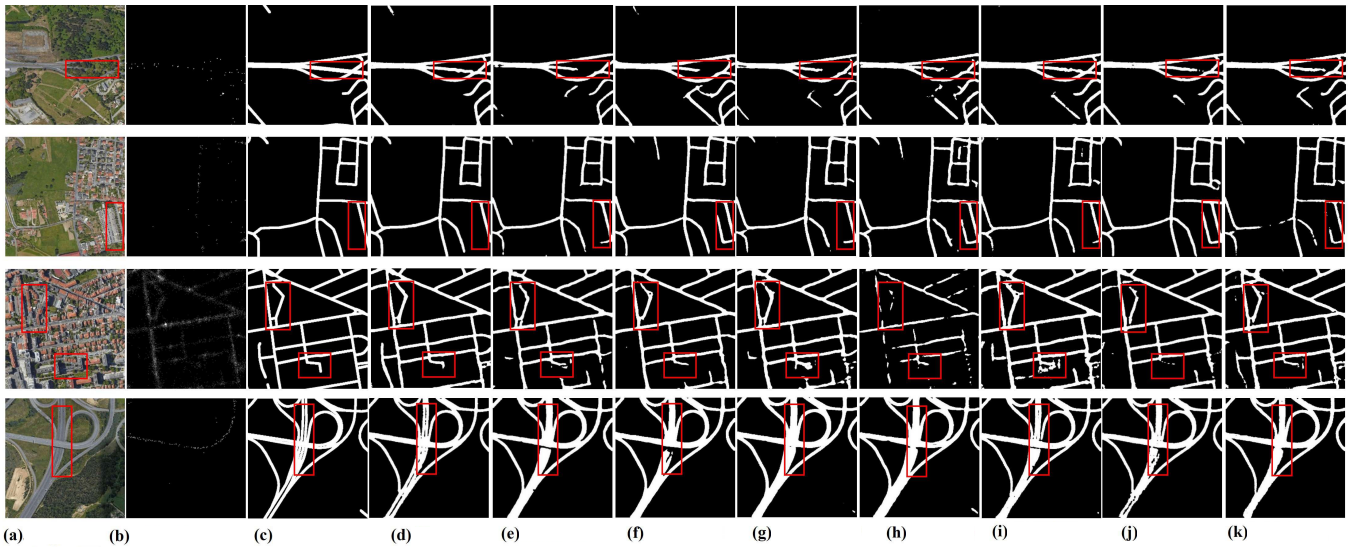


Fig. 9. Comparison of visual results among models on the Porto dataset. (a) High-resolution RS. (b) GPS trajectory. (c) Ground truth. (d) Ours. (e) CMXNet. (f) SGACNet. (g) CMFormer. (h) SA-Gate. (i) CMMPNet. (j) Sun et al. (k) DLinkNet.

TABLE III
COMPARISON OF PERFORMANCE AMONG DIFFERENT MODELS IN PORTO DATASET, WHERE BOLD NUMBERS INDICATE THE OPTIMAL VALUE FOR THIS METRIC

	IoU	F1	APLS
Ours	**0.769**	**0.870**	**0.741**
CMMPNet (TNNLS'2022)	0.747	0.849	0.725
CMIPNet (TGRS'2024)	0.735	-	0.737
SGACNet (Sensors'2023)	0.719	0.838	0.731
CMFormer (PR'2023)	0.699	0.823	0.701
CMX (TITS'2023)	0.724	0.841	0.737
DLinkNet (CVPR'2018)	0.702	0.813	0.706
Sun et al. (CVPR'2019)	0.716	0.832	0.726
SA-Gate (ECCV'2020)	0.713	0.851	0.726

features in high-resolution RS images with the corresponding road scenes represented in the GPS trajectory maps. The poor performance of SGACNet and SA-Gate may be attributed to the unsuitability for RS road scenarios and the inefficiency of their methods for cross-modal global feature aggregation between spatial and channel dimensions.

In the first row of Fig. 9, the roads extracted by other models exhibit significant serrations, which is due to the insufficient global feature learning and inadequate cross-modal feature fusion of these models. In contrast, our CFIFNet benefits from the supplementation of GPS trajectory information and enhances the differentiation between road and background features within each modality through global interaction of high-frequency information and local interaction of low-frequency information across channel and spatial dimensions. In addition, the frequency-domain adaptive filtering used in the AFCA module, which simulates the effects of large kernel convolution, further expands the scope of feature learning, thereby assisting in addressing the road extraction issue caused by severe tree obstruction, successfully extracting relatively smooth roads. In the second row, other comparative models mistakenly classify some road-like areas as actual roads, possibly because these areas have limited GPS trajectory information, preventing these models from effectively utilizing it and overly relying on high-resolution RS features for road recognition. In contrast, the CFIFNet achieves comprehensive feature complementation and fusion through the WIM structure and FFF module. Even when GPS trajectory information is scarce, it can supplement and align corresponding high-resolution RS road features. In the fourth row, when GPS trajectory data are missing, other comparative models often erroneously predict the narrow background areas between two roads as roads. This is because they fail to enhance road channel features while fully exploiting spatial information. Our model, by using the AFCA module to enhance channel features within each modality and promote channel feature interaction between modalities, successfully avoids this issue. In the third row, the roads extracted by other models are obstructed by the shadows of dense buildings, leading to incomplete prediction results. Our model, through global interaction of high-frequency information in the WIM structure and adaptive frequency-domain filtering in the AFCA module, enhances intermodal and intramodal road features, successfully extracting road edge areas and dead-end road, demonstrating higher accuracy and connectivity.

Considering that the Porto dataset primarily consists of thin road targets, a dedicated metric—Thin IoU—is designed to analyze the prediction performance of narrow road regions in this scenario. Specifically, road regions with a width of less than 3 pixels are extracted separately from the prediction and ground-truth to obtain the thin prediction region ($\text{Pred}_{\text{thin}}$) and thin ground-truth region (GT_{thin}), respectively. The Thin IoU is calculated as shown in the following equation:

$$\text{Thin IoU} = \frac{\text{Pred}_{\text{thin}} \cap \text{GT}_{\text{thin}}}{\text{Pred}_{\text{thin}} \cup \text{GT}_{\text{thin}}}. \tag{21}$$

Furthermore, the prediction and GT are processed via OpenCV to extract boundary regions (denoted as Pred_b and GT_b, respectively), aiming to focus on the prediction

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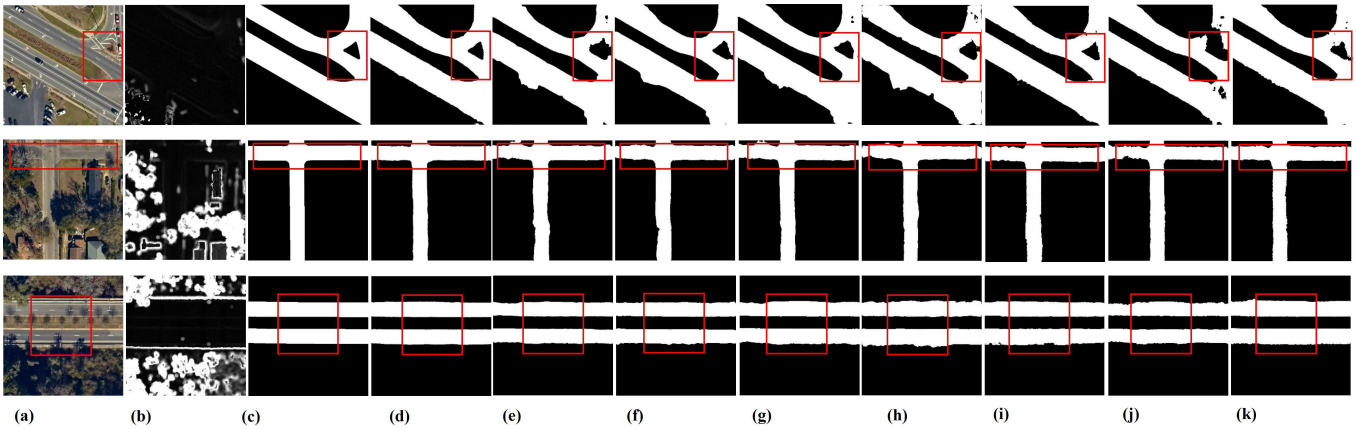


Fig. 10. Comparison of visual results among models on the TLCGIS dataset. (a) High-resolution RS. (b) LiDAR image. (c) Ground truth. (d) Ours. (e) CMXNet. (f) SGACNet. (g) CMFormer. (h) SA-Gate. (i) CMMPNet. (j) Sun et al. (k) DLinkNet.

TABLE IV
COMPARISON OF THIN IOU AND BOUNDARY IOU AMONG DIFFERENT MODELS IN PORTO DATASET

	Thin IoU	Boundary IoU
Ours	**0.728**	**0.712**
CMMPNet	0.724	0.705
SGACNet	0.721	0.698
CMFormer	0.715	0.690
CMX	0.719	0.696
DLinkNet	0.691	0.695
Sun et al.	0.699	0.702
SA-Gate	0.712	0.707

TABLE V
COMPARISON OF PERFORMANCE AMONG DIFFERENT MODELS IN TLCGIS DATASET, WHERE BOLD NUMBERS INDICATE THE OPTIMAL VALUE FOR THIS METRIC

	IoU	F1	APLS
Ours	**0.857**	**0.924**	**0.514**
CMMPNet (TNNLS'2022)	0.831	0.907	0.479
CMIPNet (TGRS'2024)	0.845	-	0.509
SGACNet (Sensors'2023)	0.829	0.906	0.473
CMFormer (PR'2023)	0.780	0.876	0.432
CMX (TITS'2023)	0.841	0.916	0.459
DLinkNet (CVPR'2018)	0.794	0.885	0.467
Sun et al. (CVPR'2019)	0.814	0.895	0.472
SA-Gate (ECCV'2020)	0.837	0.917	0.479

performance of road boundary regions. The Boundary IoU is defined as shown in the following equation:

$$\text{Boundary IoU} = \frac{\text{Pred}_b \cap \text{GT}_b}{\text{Pred}_b \cup \text{GT}_b}. \tag{22}$$

As observed in Table IV, since the performance is evaluated exclusively on the hard-to-predict regions, the performance metrics of Thin IoU and Boundary IoU exhibit a slight degradation compared with the original IoU. Nevertheless, they still outperform those of the comparative models.

3) Experimental Results on the TLCGIS Dataset: In Table V, the proposed CFIFNet outperforms CMMPNet and CMIPNet in terms of IoU by 2.6% and 1.2%, respectively. This improvement is likely attributed to the fact that, although CMMPNet and CMIPNet enable comprehensive modality information propagation during the encoding–decoding stage, some misleading information generated in the LiDAR images may persist throughout the entire process. Moreover, despite their consideration of global feature learning and information propagation, these methods mainly focus on visually salient but easily misclassified spatial features while neglecting intrinsic channel features. In contrast, the CFIFNet performs cross-modal information propagation and interaction only during the encoding stage, ensuring alignment and complementarity of low-frequency features in both spatial and channel dimensions, while the global learning capability of the Transformer guarantees the alignment and learning of high-frequency features. This design fully leverages the high resolution and rich geophysical information of RS images, as well as the strong penetration and occlusion-free characteristics of LiDAR data. Furthermore, during the decoding stage, the proposed method adopts the FFF module to fuse the features of the two modalities from a frequency-domain perspective, thereby avoiding over-reliance on any single modality. In addition, the proposed model achieves a 4.3% higher IoU compared with Sun et al. This improvement can be attributed to the fact that, although Sun et al. utilized 1-D convolutions to adapt to the distribution of road features, their method failed to sufficiently promote cross-modal feature interactions. The CFIFNet also surpasses SGACNet and CMFormer in IoU by 2.8% and 7.7%, respectively. This advantage can be explained by the fact that, although these two methods are widely recognized multimodal semantic segmentation models, they fail to adequately address the complexity of road features and RS scenes.

In Fig. 10, the second and third rows demonstrate that CFIFNet can extract relatively smooth roads, which can be attributed to its interaction between high-frequency and low-frequency regional features, effectively avoiding information entanglement. In the first row, the roads extracted by other comparative models exhibit severe erosion or dilation issues, which arise from their excessive reliance on spatial dimension learning while neglecting the enhancement of channel features. In contrast, the CFIFNet benefits from the fact that amplitude

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TABLE VI
ABLATION RESULTS OF VARIOUS MODULES IN CFIFNET ON METRIC INDICATORS

Methods	BJRoad			Porto			TLCGIS			
	IoU	F1	APLS	IoU	F1	APLS	IoU	F1	APLS	Params
Baseline	0.625	0.760	0.568	0.747	0.846	0.727	0.833	0.898	0.486	50.516
Baseline+WIM	0.634	0.769	0.574	0.755	0.854	0.738	0.841	0.909	0.497	80.806
Baseline+WIM+FFF	0.642	0.778	0.588	0.760	0.861	0.745	0.850	0.917	0.509	83.166
Baseline+WIM+FFF+AFCA	**0.650**	**0.787**	**0.594**	**0.769**	**0.870**	**0.751**	**0.857**	**0.924**	**0.517**	83.71
Baseline+WIM(CWT)+FFF+AFCA	0.642	0.783	0.588	0.762	0.865	0.742	0.851	0.919	0.509	94.65
Baseline+WIM(ML-DWT)+FFF+AFCA	0.635	0.771	0.580	0.758	0.862	0.735	0.848	0.915	0.501	94.65

reflects low-frequency global information, while phase captures high-frequency local details. The proposed FFF module achieves more comprehensive feature integration of multimodal data across multiple scales by fusing features from both amplitude and phase separately. In addition, it is observed that Sun et al. and DLinkNet simply concatenate high-resolution RS features and LiDAR features at the input stage. These methods fail to align and complement the two modalities at a deeper level, which prevents them from fully utilizing the strengths of both data modals, resulting in relatively poor prediction performance. Furthermore, we can also observe that excessive noise exists in the LiDAR images. However, through the application of modules such as WIM, FFF, and AFCA, our model effectively mitigates the impact of noise.

V. ABLATION STUDY

A. Ablation of Modules

From Table VI, it can be observed that when WIM is incorporated into the encoder part of the baseline model, there is a significant improvement in performance metrics, reaching 0.9%, 0.9%, and 1.0% on $F1$, IoU, and APLS in the BJRoad dataset, respectively. This is attributed to its ability to align features of both modalities in the encoder while learning features from each other's modality data. In addition, the WIM interacts with the high-frequency and low-frequency information of both modalities separately, avoiding the learning of irrelevant information from the other modality and ensuring the extraction of high accuracy and connectivity of road features. When FFF is added to Baseline + WIM, there is a slight performance improvement, reaching 0.8%, 0.5%, and 0.9% in the $F1$ on BJRoad, Porto, and TLCGIS datasets, respectively, which is due to its lightweight approach to fuse amplitude and phase values from both modalities, achieving a more informative representation of frequency-domain signals. Finally, the addition of the AFCA module to Baseline + WIM + FFF further enhances model performance, accompanied by an increase of 0.9%, 0.9%, and 0.7% in the $F1$ on BJRoad, Porto, and TLCGIS dataset, respectively, which is attributed to its enhancement of the distinctiveness between high-resolution RS and GPS trajectory/LiDAR feature channels, as well as the expansion of the receptive field during feature learning. Thus, the prevalent occlusion issues at urban and rural road edges are effectively alleviated, ensuring that the pixel-level classification performance of these regions is improved. Meanwhile, we replaced the 2-D DWT in the WIM with multilevel DWT (ML-DWT) and continuous wavelet transform (CWT), respectively. It was found that the performance slightly degraded.

TABLE VII
ABLATION RESULTS OF AFCA, AFSCA, AND DEFORMABLE ATTENTION IN BJROAD DATASET

Modules	Params	IoU	F1	APLS
Deformable Attention	118.55	**0.658**	**0.794**	**0.605**
AFCA	**83.71**	0.650	0.787	0.594
AFSCA	83.71	0.654	0.790	0.598

This may be attributed to the fact that ML-DWT performs repeated transformations based on low-frequency components, leading to insufficient utilization of high-frequency components; for CWT, the low-frequency components across continuous scales exhibit redundancy, which interferes with subsequent feature fusion processes.

To highlight the superior performance of AFCA, we conduct a comprehensive comparison with the widely adopted deformable attention module [52], evaluating both the parameter efficiency and performance when integrated into the model. As shown in Table VII, the results reveal that while AFCA exhibits marginally lower performance metrics compared with deformable attention, it achieves a substantial reduction in parameter count. Furthermore, we replaced the channel attention component in the AFCA module with frequency–spectral channel attention [53], constituting the adaptive frequency–spectral convolution attention (AFSCA) module. It was found that the number of parameters remained unchanged, while the performance was slightly improved. This is because, compared with AFCA, AFSCA performs compression by selecting multiple discrete cosine transform (DCT) frequency components, embedding richer information into the scalar representation of each channel. In contrast, AFCA directly uses GAP—which corresponds to only the lowest frequency information—to represent channel information.

B. Ablation of Vision

From Fig. 11, along with using the Grad-CAM [54] method to observe the visual heatmap results from the last convolution layer of the model, it can be observed that with the addition of trajectory/LiDAR information, the predicted road confidence in complex scenes is higher, which is also attributed to the efficient utilization of multisource information by modules such as WIM, FFF, and AFCA. It can also be seen that trajectory information can enhance the prediction effect on small target roads that are easily overlooked and roads obscured by trees or buildings, while LiDAR maps can increase the confidence of road edges. Specifically, when the model does not include the WIM structure, the overall prediction confidence is low,

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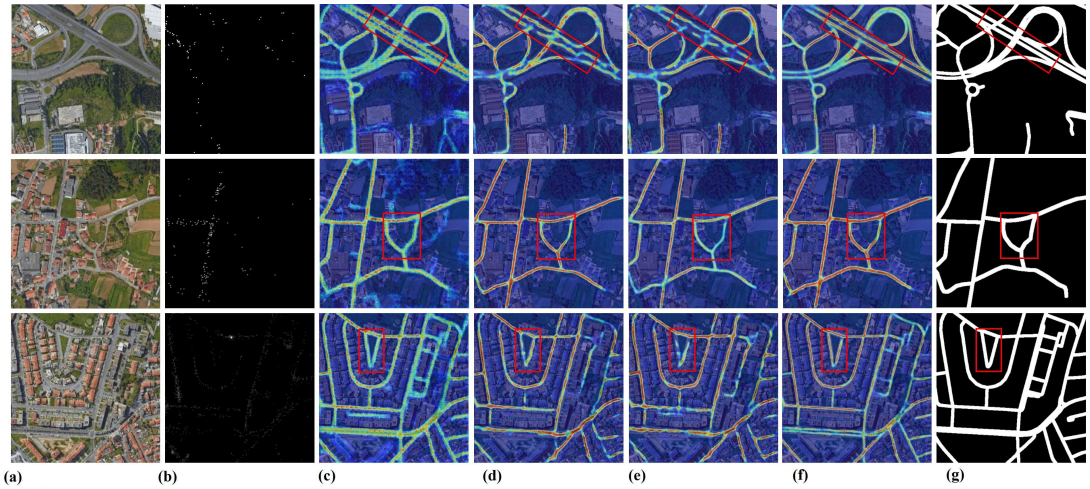


Fig. 11. Visual ablation results in the last convolutional layer of the model for WIM, FFF, and AFCA, where darker colors indicate higher confidence. (a) High-resolution RS. (b) GPS trajectory. (c) Without WIM. (d) Without FFF. (e) Without AFCA. (f) WIM + FFF + AFCA. (g) Ground truth.

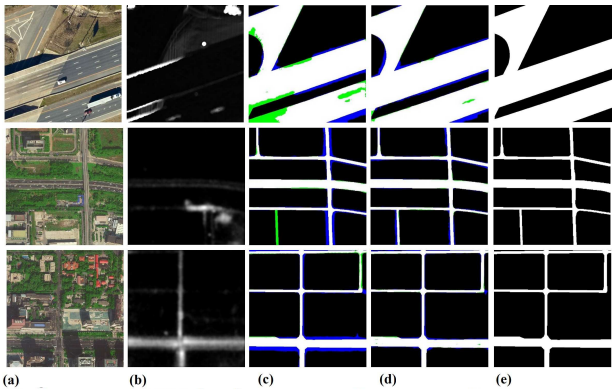


Fig. 12. Comparison of visual results before and after adding WIM. The blue regions indicate pixels where the ground-truth is background, but are predicted as road. The green regions indicate pixels where the ground-truth is road but are predicted as background. The white regions indicate pixels where both ground-truth and prediction are road. (a) High-resolution RS. (b) GPS trajectory or LiDAR. (c) Before adding WIM. (d) After adding WIM. (e) Ground truth.

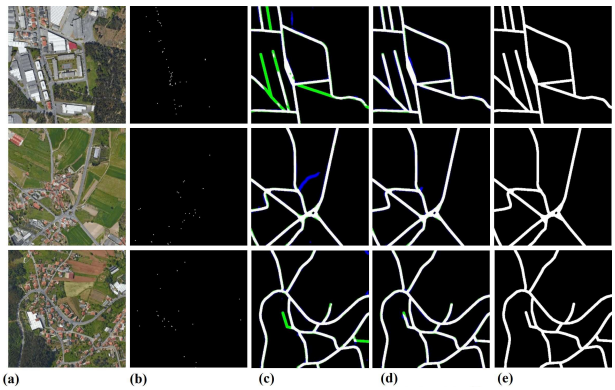


Fig. 13. Comparison of visual results before and after adding FFF. (a) High-resolution RS. (b) GPS trajectory or LiDAR. (c) Before adding FFF. (d) After adding FFF. (e) Ground truth.

possibly due to insufficient alignment and interaction of features from the two modalities, as well as excessive intermodal noise. When the model does not include the FFF module, the extracted roads exhibit severe serrations, possibly due to insufficient fusion of modal information and the interference from redundant information. When the model does not include the AFCA module, some road edges in densely built areas are incorrectly predicted as the background, which may be attributed to insufficient long-range feature learning and inadequate enhancement of road features. At the same time, although GPS trajectory information or LiDAR information may differ significantly from the ground-truth of roads, due to the use of WIM, FFF, and AFCA modules in our model, features from the GPS trajectory branch or LiDAR branch can align with features in the high-resolution RS branch, avoiding interference from incorrect or invalid information in the final prediction stage, and further increasing the difference between background and foreground road features.

As seen in Fig. 12, before adding the WIM to the model, a large number of road edge areas were misclassified as background, and simultaneously, many edge background areas were misclassified as roads. This might be due to the high similarity of foreground–background features at the boundary positions. It could also be that certain GPS trajectory information deviated from the high-resolution RS scenario, leading to misjudgments in the road edge areas. When the WIM is added to the model, the prediction performance of road features in the edge regions improves, with the extraction of more accurate road edge contours. This may be attributed to the alignment of high-frequency and low-frequency features between the two modalities through the WIM. It preserves the overall structure of the road, provides a macro distribution of the road, and enhances the performance of road skeleton extraction through the interaction of low-frequency information, while increasing the difference between road features and surrounding background features, thus avoiding the appearance of jagged effects through the alignment of high-frequency information.

As observed in Fig. 13, the model without the FFF module contains some false road predictions, where some rural paths

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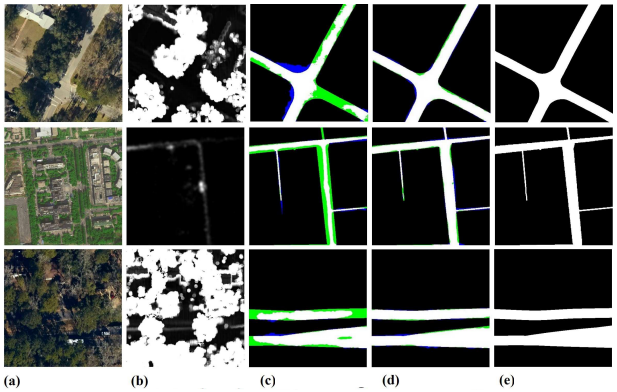


Fig. 14. Comparison of visual results before and after adding AFCA. (a) High-resolution RS. (b) GPS trajectory or LiDAR. (c) Before adding AFCA. (d) After adding AFCA. (e) Ground truth.

that were not labeled as roads are mistakenly identified as roads. This could be attributed to the fact that these areas have textures and colors similar to those of traffic roads, leading to misclassification. In addition, there is significant noise interference in the GPS trajectory/LiDAR maps. During the feature downsampling and abstraction learning process in the encoding stage, the features from the two modalities were not fully complementary or aligned, which may have resulted in inaccurate predictions of narrow and small target roads. However, after adding the FFF module, the model efficiently avoids false road predictions and accurately extracts narrow and small target roads. This improvement can be attributed to the FFF module's ability to retain spatial structural information, mitigate noise interference, and highlight the significant features of the two data sources through separate fusion and filtering of magnitude and phase.

As shown in Fig. 14, the model without the AFCA module performs poorly in urban scenes where buildings and trees cause occlusion and rural scenes covered by forests. This may be due to information loss during the upsampling process and the inaccuracy of the GPS trajectory/LiDAR maps. However, after adding the AFCA module, the model's visualization performance is further improved in the occluded scenarios mentioned above. This improvement can be attributed to the AFCA module, which enhances road features through the learning of global features from the frequency-domain perspective and the interaction of global features in the channel dimension. In addition, when GPS trajectory/LiDAR information is insufficient or shifted, the AFCA module resolves edge occlusion issues through lightweight long-range dependencies to maximize the utilization of RS data.

C. Ablation of Special Scenes

Due to the absence of foggy scenes in the aforementioned three multisource datasets, inspired by the classic multisource data road extraction method CMMPNet [1], we employed the fog effect renderer in the PhotoShop tool to generate multiple foggy high-resolution images, with the aim of increasing the training sample size and enhancing the model's adaptability to complex scenarios. As shown in Fig. 15, the prediction results

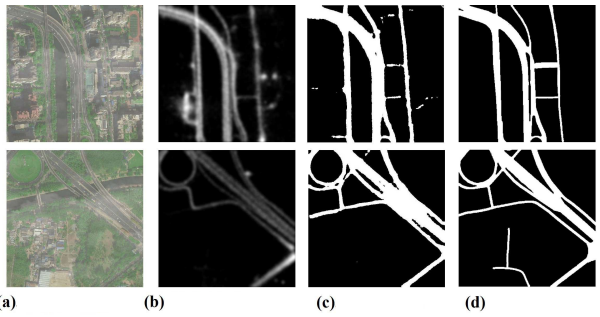


Fig. 15. Visualization results of high-resolution RS images with added fog effects. (a) Fogged high-resolution RS. (b) GPS trajectories. (c) Predictions. (d) Ground truth.

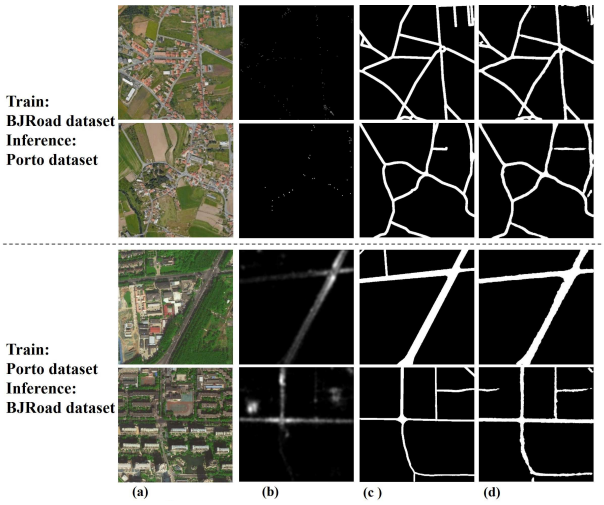


Fig. 16. Visualization results of cross validation. (a) High-resolution RS. (b) GPS trajectory. (c) Ground truth. (d) Predictions.

TABLE VIII
ABLATION RESULTS OF CROSS VALIDATION IN BJROAD AND PORTO DATASETS

Strategy	IoU	F1
Train:BJRoad, Inference:Porto	0.735	0.832
Train:Porto, Inference:BJRoad	0.616	0.749

in foggy weather scenarios indicate that even though some information in high-resolution RS images becomes blurred due to fog obstruction, the CFIFNet can still predict high-quality road structure maps. This is attributed to the efficient utilization and supplementation of GPS trajectory information by modules such as WIM, FFF, and AFCA, which compensate for the limitations of RS images in such scenarios, with effectively avoiding intermodal noise interference and suppressing intramodal noise.

D. Cross Validation for Generalization Ability

To validate the generalization capability of our CFIFNet for cross-city multisource RS images feature alignment, we employ a cross-validation strategy: training on the BJRoad dataset while testing on the Porto dataset, and vice versa. As previously noted in Section IV, BJRoad and Porto datasets

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exhibit significant differences in road occlusion environments, road scales, and GPS trajectory information richness. Consequently, sharing identical model parameters across both datasets presents significant challenges due to their inherent domain gaps. Nevertheless, as demonstrated in Table VIII and Fig. 16, CFIFNet maintains robust cross-domain performance, reducing the interdomain performance gaps.

VI. Conclusion and Outlook

In this study, we propose the CFIFNet based on frequency-domain learning, a novel framework that addresses key challenges in multisource road extraction, which incorporates two independent encoding branches to separately process high-resolution RS data and GPS trajectory/LiDAR information. In addition, a WIM structure is designed to interact and complement the high-frequency and low-frequency features of both modalities during the encoding stage, effectively aligning intermodal edge features and enabling accurate extraction of road edges with high background similarity. Furthermore, a FFF module is introduced to fuse complementary information of both modalities from both amplitude and phase perspectives, enhancing the distinction between roads and background, thereby preventing the loss of small road segments and misclassification of objects similar to roads. Considering the issue of road edge occlusion, an AFCA structure is designed to expand the receptive field from both channel- and frequency-domain dimensions. Extensive comparative experiments on multiple multimodal road datasets demonstrate the effectiveness of our model over other cross-modal road extraction approaches. Moreover, a series of ablation studies validates the specific contributions of the structures and methods adopted in CFIFNet, as well as their applicability and generalization capabilities.

In addition, although our model achieves promising results across multiple multimodal road datasets, we must acknowledge its strong reliance on high-quality annotated data. It is also worth noting that annotating multimodal data is more challenging and prone to errors compared with annotating high-resolution RS data alone. Therefore, in future research, we plan to adopt a semi-supervised approach by annotating a small subset of samples within multimodal datasets to explore their intrinsic consistency of road features with other unlabeled samples.

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